Week 99: Moving piles

Adult guide built from the problem files · Unpiloted; physically unrehearsed.

The answer keys, proofs, preparation and launch are in the print guide, week-99-facilitator.pdf, whose source is guide/facilitator.tex and guide/solutions.tex. This view lists each problem's numbers, route and idea.

Problems and routes

Problem	2–3	4–5	route
Match each target with the least work	P1	P1	first
Least work for three pairs	P2	P2	first
Does the shortest delivery first always work?	P3	P3	first
Starts with least work 4 and 7 to a fixed target	P4		return
Eight arrangements with position total 8		P4	return
Moving counters on star roads	P5	P5	return
A start at least work exactly 5	P6		return
Can a detour through B beat going straight from A to C?		P6	return

First route. Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** Grades 2–3 P4–P6; Grades 4–5 P4–P6.

Key, one entry per problem

Match each target with the least work

Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Each gap between bins must be crossed by a fixed net number of counters.

Least work for three pairs

Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. The gap counts bound every plan, and matching counters in order meets the bound.

Does the shortest delivery first always work?

Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. Making the shortest delivery first can fail: 0, 3 to 2, 5 costs 6 that way and 4 by matching in order.

Starts with least work 4 and 7 to a fixed target

Grades 2–3 P4 · return route

Idea. With every counter going to bin 2, the least work is the sum of distances to 2.

Eight arrangements with position total 8

Grades 4–5 P4 · return route

Idea. Equal position totals can still be 8 apart: 0044 and 2222.

Moving counters on star roads

Grades 2–3 P5 · Grades 4–5 P5 · return route · wording differs by band

Idea. On a star each road's imbalance must cross it, and the sum of those imbalances is the least work.

A start at least work exactly 5

Grades 2–3 P6 · return route

Idea. Every target has a start at least work exactly 5, built by five moves in one direction.

Can a detour through B beat going straight from A to C?

Grades 4–5 P6 · return route

Idea. Least work obeys the triangle rule: D(A, C) is at most D(A, B) + D(B, C).